UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

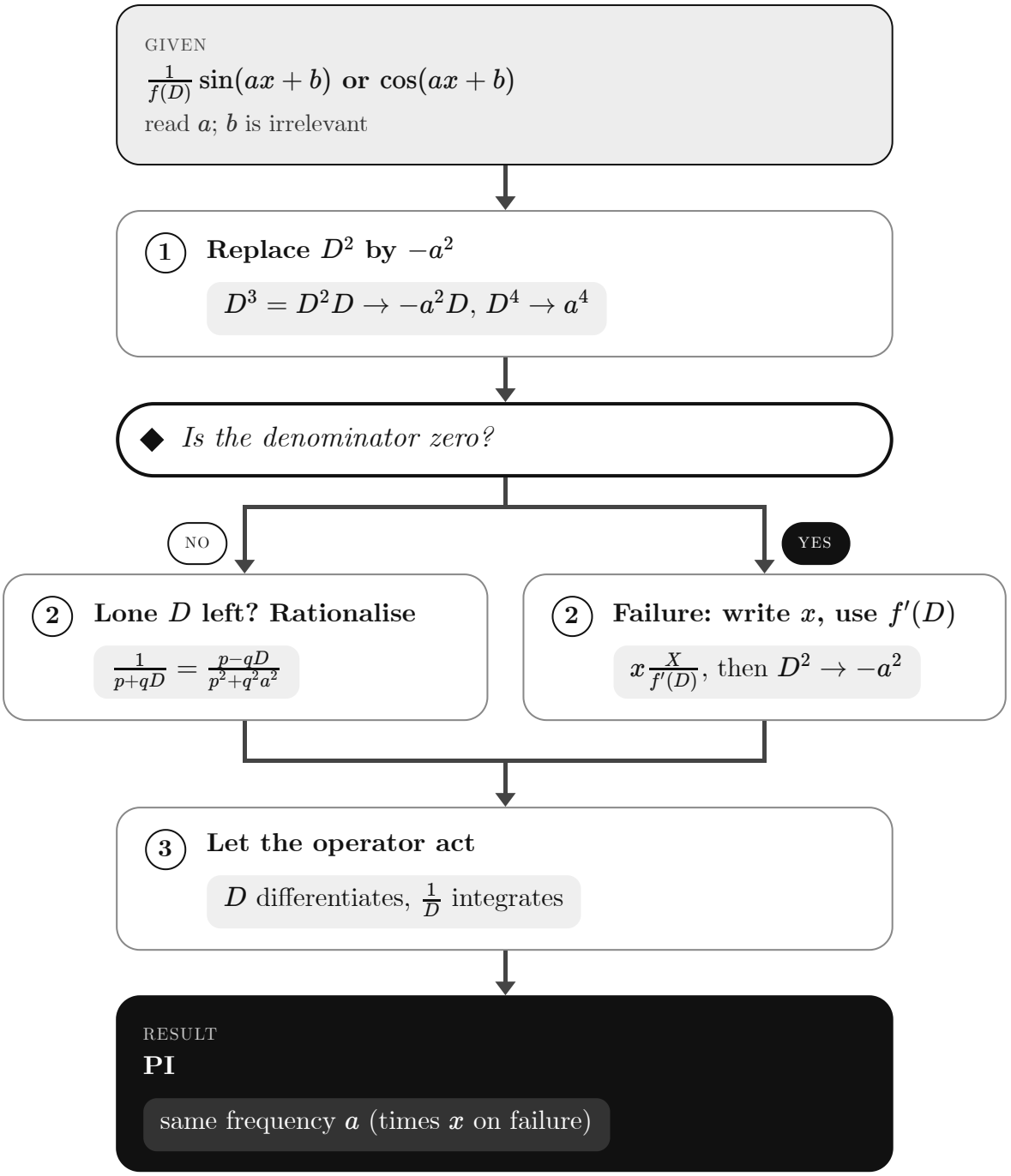
Class 004 · Particular integral Type 2 (sin ax, cos ax) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9	Unit 2 deck 004 (“Particular integral of Type 2”), slide 9 (rendered .pptx)
T004 00:10:32	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 2” (24.7 min), segment start
T004b 00:08:13	Unit 2 AV-summary transcript “Particular integral Type 1 and Type 2” (12.6 min, newer recording); its Type 1 part is tagged T004a in the class 003 pack
CW U2 P13	Unit 2 class-work problem 13
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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1 The Type 2 rule: replace D^2 by $-a^2$



Remark. The CF warns you: if $\pm ai$ are AE roots, the answer is "yes".

Exam trap. $-a^2$, not $(-a)^2$.

Figure F1.1. The Type 2 fork: does $f(-a^2)$ vanish? ESA ★★★

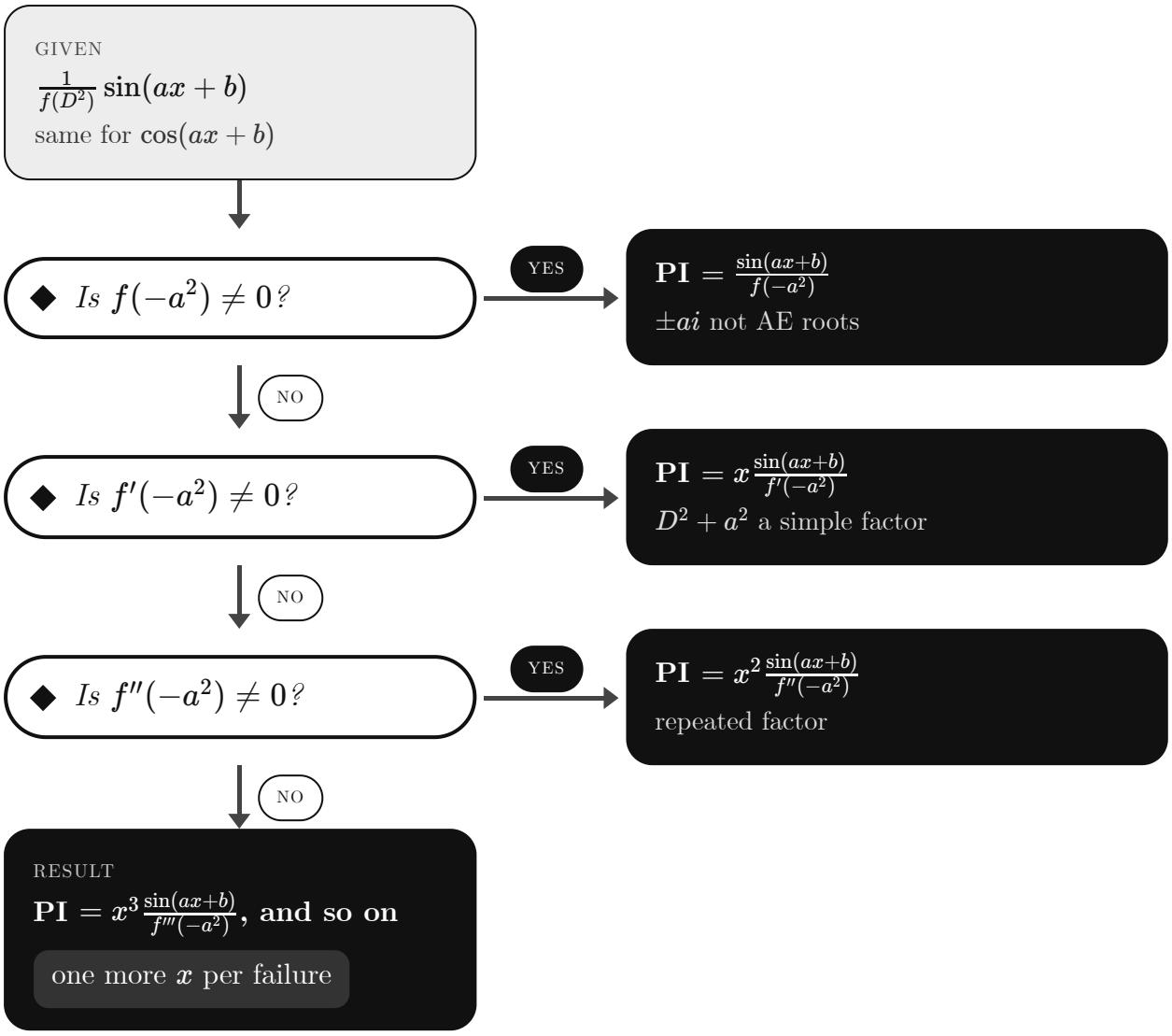
S004·4 · S004·5 · T004 00:01:15 · T004 00:11:48 · T004b 00:07:32

Type 1 and Type 2 side by side

	Type 1: $X = e^{ax}$	**Type 2: $X = \sin(ax+b)$, $\cos(ax+b)$**
Replace	D by a	D^2 by $-a^2$ (odd D kept, then rationalised)
PI (no failure)	$\frac{e^{ax}}{f(a)}$	$\frac{X}{f(-a^2)}$
Fails when	a is an AE root	$\pm ai$ are AE roots ($D^2 + a^2$ divides f)
First failure	$x\frac{e^{ax}}{f'(a)}$	$x\frac{X}{f'(-a^2)}$
Second failure	$x^2\frac{e^{ax}}{f''(a)}$	$x^2\frac{X}{f''(-a^2)}$

S004·4 · S004·5 · T004b 00:00:41 · T004b 00:06:12

2 Case of failure: $f(-a^2) = 0$

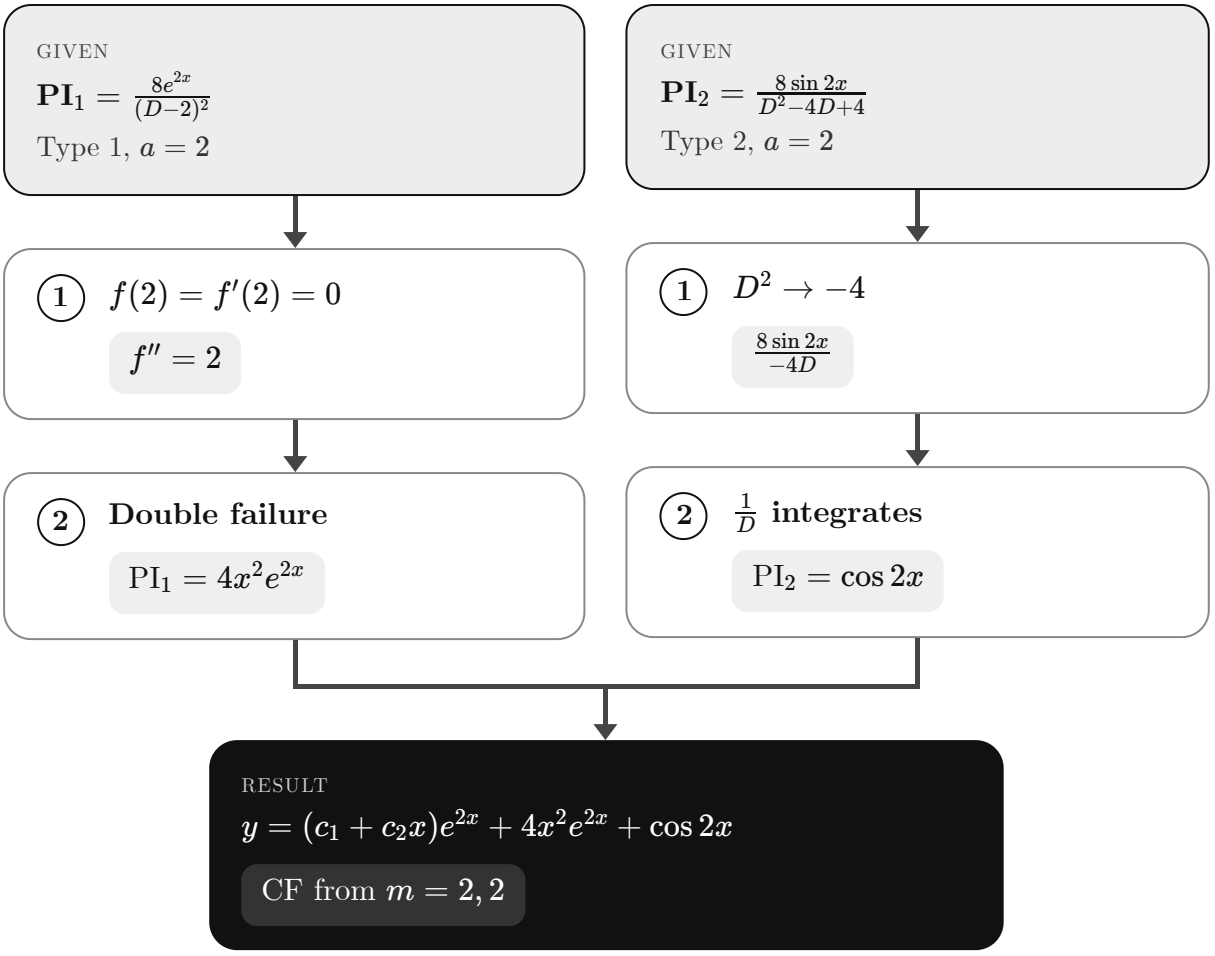


Remark. Derivatives are with respect to D; any odd D left is rationalised or, if alone, integrated ($\frac{1}{D}$).

Exam trap. Slide 004·15 writes $f'(a)$; read $f'(-a^2)$.

Figure F2.1. The Type 2 ladder ESA ★★★

S004·4 · S004·5 · S004·6 · S004·15 · T004 00:04:04 · T004 00:05:06

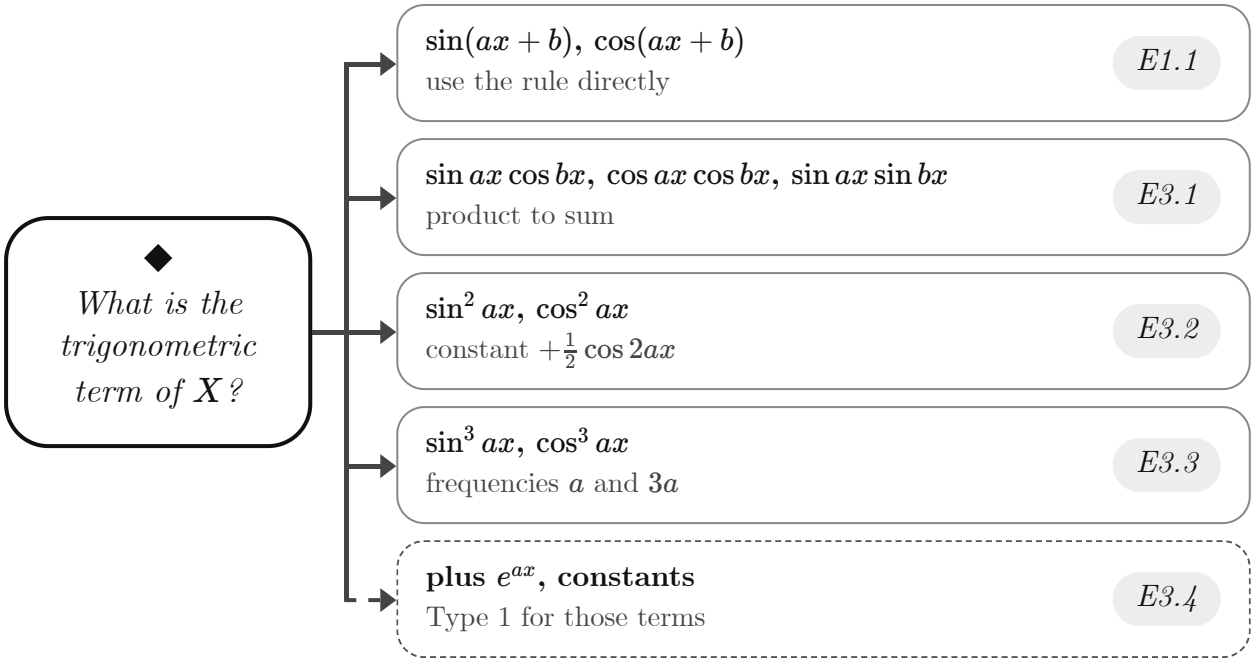


Remark. No rationalising is needed: the denominator is a pure D, so $\frac{1}{-4D}$ is just $-\frac{1}{4}\int$.

Figure F2.2. ESA: $(D-2)^2y = 8(e^{2x} + \sin 2x)$ ESA ★★★

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

3 Products, powers and sums: reducing to Type 2



Remark. After rewriting, every term is either Type 1 ($D \to a$) or Type 2 ($D^2 \to -a^2$); add the PIs.

Figure F3.1. Reduce X to Type 2 first ESA ★★★

S004·7 · T004 00:05:24 · T004 00:06:12 · T004 00:20:36

Identities that reduce X to Type 2

X	Rewrite as	Rule for each term
$\sin ax \cos bx$	$\frac{1}{2}[\sin(a+b)x + \sin(a-b)x]$	Type 2 at $a+b$ and $a-b$
$\cos ax \cos bx$	$\frac{1}{2}[\cos(a+b)x + \cos(a-b)x]$	Type 2 at $a+b$ and $a-b$
$\sin ax \sin bx$	$\frac{1}{2}[\cos(a-b)x - \cos(a+b)x]$	Type 2 at $a-b$ and $a+b$
$\sin^2 ax$	$\frac{1}{2} - \frac{1}{2}\cos 2ax$	Type 1 ($D \to 0$) + Type 2 at $2a$
$\cos^2 ax$	$\frac{1}{2} + \frac{1}{2}\cos 2ax$	Type 1 ($D \to 0$) + Type 2 at $2a$
$\sin^3 ax$	$\frac{3}{4}\sin ax - \frac{1}{4}\sin 3ax$	Type 2 at a and $3a$
$\cos^3 ax$	$\frac{3}{4}\cos ax + \frac{1}{4}\cos 3ax$	Type 2 at a and $3a$

ADDED FOR CLARITY S004·7 · T004 00:05:42 · T004 00:06:23

The Type 2 term inside every ESA question

Paper, Q	$f(D)$	Type 2 term	Its PI	Full solution
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(D-2)^2$	$8\sin 2x$	$\cos 2x$ (W2.3)	here, W2.3
Jul-23 3b	D^2-5D+6	$\sin x$	$\frac{\sin x+\cos x}{10}$ (W1.3)	class 005
Jan-24 s1 2d	D^2+4	$\cos 2x$	$\frac{x\sin 2x}{4}$, failure (W2.2)	class 005
Jan-24 s3 2c	D^3-2D^2+D	$\sin^2 x$	$\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$ (W3.2)	class 006
Jan-24 s4 2c	$(D-1)^2(D+1)^2$	$\sin^2\frac{x}{2}$	$\frac{1}{2}-\frac{\cos x}{8}$ (W3.5)	class 005

ADDED FOR CLARITY

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c